MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
First Semester

Multi-Disciplinary Course - MG1MDCCSC101 - INTERNET AND WEB TECHNOLOGIES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour Maximum Marks: 35

Scheme for Evaluation

Part A: Very Short Answer Questions (Answer all) - (10 * 1 = 10 Marks)

1. **State any 2 roles of internet in entertainments?** [K] [1]

   Answer: (0.5 marks each)

Online Streaming –watch movies, web series, TV shows,YouTube, Netflix, Hotstar, etc.

Online Gaming –multiplayer games, cloud gaming, and interactive gaming etc.

2. **Identify the default behavior of a marquee in terms of direction.** [U] [2]

   Answer:

   The default behavior of a **<marquee>** tag **in terms of direction is:** It scrolls the text from right to left.

3. **List one key benefit of resource sharing in a network** [U] [1]

   Answer:

- Easy Communication
- Centralized Management
- Improved Collaboration

4. **Describe the term WWW.** [U] [1]

   Answer: World Wide Web

5. **Name any two attributes that can be added on video element in HTML.** [K] [2]
   Answer: Any two (0.5 marks each)

- Controls
- Autoplay
- Height
- Width
- Src

6. **List the potential disadvantages of excessive internet use** [K] [1]

   Answer: Any two (0.5 marks each)

   - Health issues
   - Addiction
   - Privacy & Security
   - Reduced Productivity

7. **Which tag was used to group multiple frames together in HTML?** [U] [2]

   Answer: <frameset>

8. **Identify the use of <code> tag.** [U] [2]

   Answer: Display a piece of computer code or programming text in a monospace (fixed-width) font. Helps display **programming code**, commands, or technical text.

9. **State the role of HTTP in internet working** [U] [1]

   Answer:

HTTP (HyperText Transfer Protocol) is responsible for **transferring web pages and other resources between a web server and a web browser**.

It defines how requests and responses are sent, enabling users to access websites on the Internet.

10. **Which attribute of the <textarea> tag defines the width of the text area.(U)[2]**

Answer: The attribute of the **<textarea>** tag that defines the **width** of the text area is:**cols**

Part B: Short Answer Questions (Answer any 3 out of 5 Questions) - (3 * 5 = 15 Marks)

11. **Differentiate between micro wave and radio wave.** [K] [1]

Answer: (2.5 marks for comparison of each factor)

- Microwaves are electromagnetic waves which have frequency range between 1 GHz to 300 GHz. These can travel along long distances. These are unidirectional in nature which means that they can travel only in straight line. At very high frequency that cannot penetrate into walls.

- The radio waves have frequency range from 3 KHz to 1 GHz.These waves are easy to generate and these can travel along long distances.
- These waves are omni directional in nature which means that they can travel in all the directions.
- They are widely used for the communication between both indoor and outdoor because they have the property that they can penetrate through the walls very easily.

12. **Explain the differences between star and ring topologies in terms of data flow and failure impact.** [U] [1]

Answer: (Figure 2 marks)

Data Flow

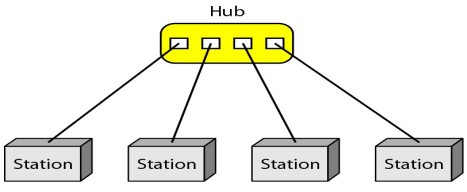
Data is sent from each device to a **central hub or switch**, which then forwards it to the destination.(star)

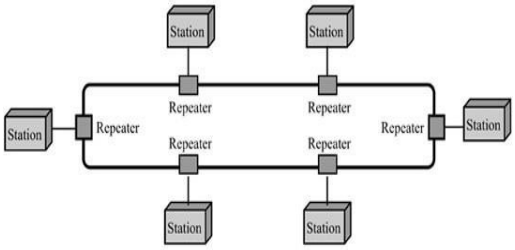
Data travels in a **circular path (ring)** from one device to the next in one direction.(ring).

Failure Impact

If one device fails, the network continues working. But if the **hub fails**, the entire network stops.(star)

Failure of **any single device or connection** breaks the ring and can stop the whole network.(ring).





13. **Discuss the importance of closing tags in HTML with an example.** [U] [2]

Answer: Closing tags play a crucial role in defining the **start and end** of HTML elements. They ensure that the structure of the webpage is correct and that elements do not overlap or behave unexpectedly. Many tags such as <p>, , <h1>, <table>, etc., **must** be closed.

14. **Discuss the benefits and drawbacks of using named colors (like “red” or “blue”) versus hexadecimal codes in HTML.** [U] [2]

Answer 1. Named Colors (e.g., “red”, “blue”, “green”)

Benefits

- Easy to remember and use — simple names make coding faster.
- Readable — improves code clarity, especially for beginners.
- No need to know exact color values — quick for basic styling.

Drawbacks

- Limited choices — HTML supports only around 140 named colors.
- Not precise — cannot represent specific shades or custom tones.
- Less professional — not suitable for detailed or brand-specific design.

2. Hexadecimal Colors (e.g., #FF0000, #0000FF)

Benefits

- Precise color control — allows millions of color variations.
- Widely used in professional web design — supports accurate branding.
- Consistent across all browsers — exact same shade everywhere.

Drawbacks

- Harder to remember — hex codes are not intuitive.
- Less readable — beginners find codes confusing.
- Takes more time to write compared to short color names.

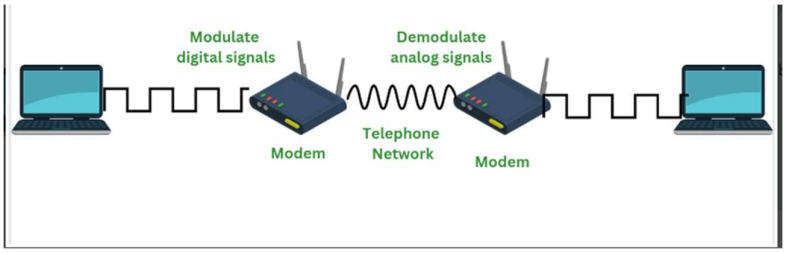
15. **Explain the operation of modem.** [K] [1]

Answer: A **modem (modulator–demodulator)** is a device that enables communication between computers over telephone lines or other analog communication systems.

Modulation = Digital → Analog

Demodulation = Analog → Digital

Modem allows long-distance data communication over analog networks.



Part C: Essay Questions (Answer any 1 out of 2 Questions) - (1 * 10 = 10 Marks)

16. **Define network security. Explain the main goal of network security.** [K] [1]
    - **Answer: Network security** refers to the set of policies, tools, and techniques used to protect a computer network from unauthorized access, misuse, attacks, and data loss. It ensures that data and network resources remain safe and available only to authorized users.
    - Main Goals of Network Security- Confidentiality, Integrity, Availability
17. **Define a hyperlink in HTML. Analyze the difference between internal and external links. Explain the role of the href attribute in creating a hyperlink.** [U] [2]
    - **Answer:** A **hyperlink** is an HTML element that allows users to move from one webpage to another or jump to a specific section within the same page. It is created using the <a> (anchor) tag.
    - Click Here
    - Internal Links -Connect one page of a website to another page **within the same website**.
    - External Links- Point to a webpage **outside the current website** (different domain).